

Unmasking Disparities: Examining the Impact of Race and Gender on Emergency Room Treatment for Young Adults

Zachary Serby and Irene Agusti

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Abstract

This study delves into a decade's worth of ER visit data from the Consumer Safety Commission to investigate the effect of race and gender-based disparities in emergency room (ER) treatment, utilizing statistical significance tests, diverse models, and visualizations. We focus on six outcomes across seven races/ethnicities and two genders. These data were modeled by a multinomial logistic regression which demonstrated statistical significance in disparate treatments for race and gender minorities. These findings contribute to the discourse on healthcare disparities in high-stakes ER settings, guiding efforts towards equitable healthcare delivery.

Introduction

“Health equity is the attainment of the highest level of health for all people.”¹ Embracing this principle, our study investigates disparities in emergency room (ER) treatment, a vital facet of healthcare delivery, which despite government acknowledgment is still a prevailing issue in the United States. Inspired by J. Blanchard's research prompt on “Racial and Ethnic Disparities in Health: An Emergency Medicine Perspective,” our goal is to research the predictors of disparities, which could help mitigate their impact in the long run. We aim to answer the question: “Do race and gender contribute to disparities in ER settings? If so, in what way?”

Data Description

In our study, we accessed the National Electronic Injury Surveillance System (NEISS) dataset through the United States Consumer Product Safety Commission website. NEISS, operated by the Consumer Product Safety Commission for over 45 years, gathers data on consumer product-related injuries from a nationally representative sample of hospitals in the U.S. This data is crucial for estimating nationwide product-related injury rates and serves as a vital tool for public health research globally. Important to note, we treat individuals in the dataset as distinct entities, not merely as a representation of hospitals with assigned weights and Primary Sampling Units (PSU).

Data Cleaning

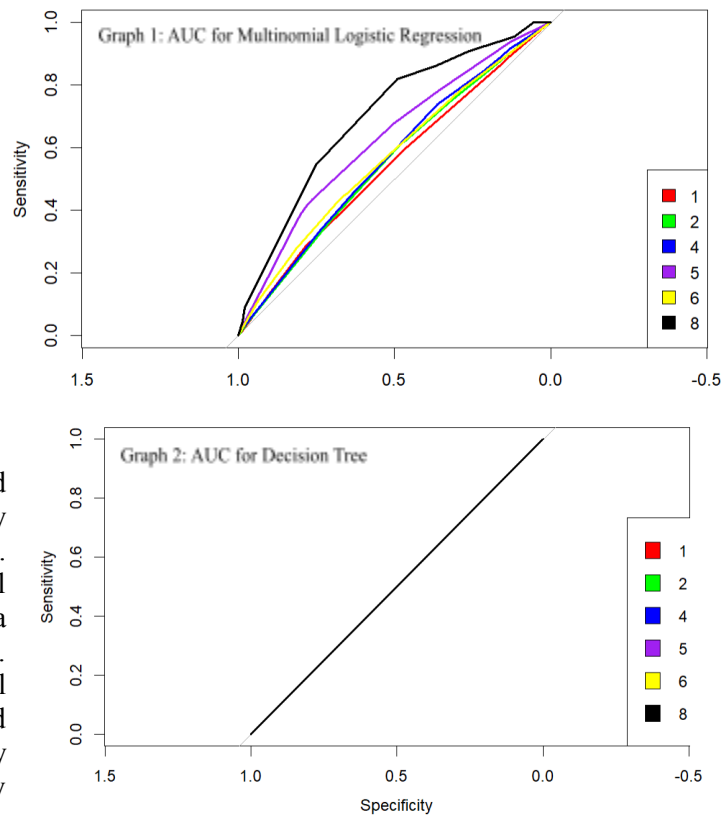
We utilized the NEISS Estimates Query Builder to extract relevant data on serious injuries (fractures, hemorrhage, lacerations, concussions, etc.) excluding dental injuries. Our focus was on young adults aged 16-25 from 2013-2022, encompassing all body parts, races, and sexes. We specifically examined the variables: race, gender, and treatment outcomes.

For data cleaning, we uploaded a zipped folder in R, iterated through each Excel file using a loop, and used the `read_excel` function to process and append the files. We removed unnecessary columns, including those with excessive NAs, singular occurrences, or pertaining to sample weights for certain hospital types. We converted relevant variables (Race, Sex, Disposition) to factors and used the `relevel` function to set the reference to Race = “White”. The key was reformatted to include underscores in column names, and variables related to removed dataset elements were excluded.

¹(U.S. Department of Health and Human Services)

Methodology

We conducted chi-squared and Fisher tests to assess the significance of sex and race (predictor variables) on disposition (response variable). The p-values for sex and race were highly significant (Sex p-value < 2.2e-16, Race p-value 0.0004998). The chi-squared test indicated a significant association between Race|Sex and treatment. For analyzing the impact of categorical variables (Race and Sex) on "dispositions," we initially used linear regression, k-nearest neighbors, and decision trees with factorized categorical variables. We also explored logistic regression, creating binary outcome columns for each treatment type. However, considering the categorical nature of our variables, we opted for a multinomial logistic regression model. Surprisingly, all models (multinomial logistic regression, decision tree, and random forest) showed identical accuracy (0.9023). We evaluated model efficacy using AUC values, finding that the multinomial logistic regression model surpassed both the decision tree and random forest models. The multinomial regression achieved an average AUC value of 0.5930 across dispositions, outperforming the others with an average AUC value of 0.5². Hence, we can confidently assert that our multinomial logistic regression model is the optimal choice for representing our data in this project.



Results

The coefficients describing race in the study revealed systematic treatment disparities across all racial categories, including gender distinctions³. For instance, when comparing black individuals to white counterparts, there's a significant decrease in the likelihood of being treated and transferred. The multinomial logistic regression analysis indicates statistically significant p-values for Native Hawaiian/Pacific Islander, American Indian/Alaska Native, black, and female gender. The visualization highlights a significant p-value⁴, suggesting the need for further investigation into the disparities between Native Hawaiian/Pacific Islander and white individuals who have died in the ER.

Discussion

Of all the models tried, multinomial regression proved to be the most efficient in terms of displaying the impact of sex and gender on the probability of certain treatment outcomes; even though the model still

² See Appendix A

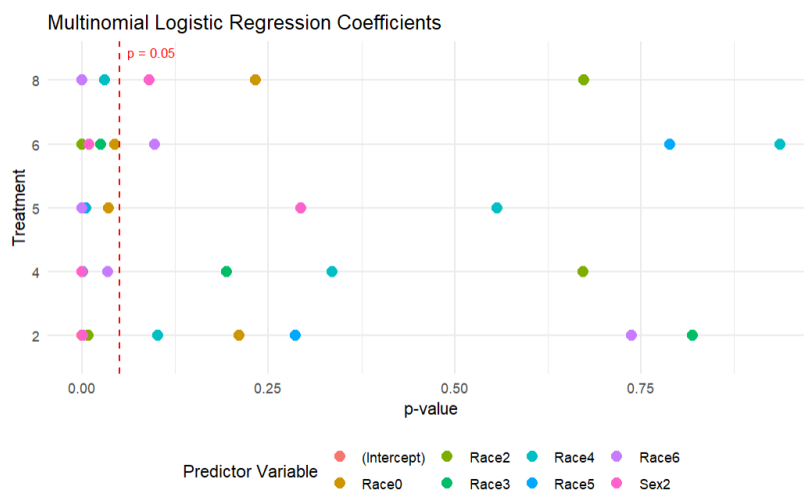
³See Appenndix A.1

⁴See Appendix A.1, purple marker.

“preferred” treatment 1, it did more accurately display the different probabilities of getting that treatment depending on race and gender based of the baseline “white male”, therefore aligning more with the results of our Chi-Square and Fisher tests. Also, given that the data came from a stratified sample with weights, it would be interesting to see whether treating it as such would make a difference with the given results. Due to the time constraints we would have liked to do models that interacted with variables, took diagnosis into consideration (either as part of the model or by making models on separate subsets) and do case by case comparison. We did do a multinomial regression to study the interactions of the variables where treatment 1 (given its dominance) was not considered, however since that implied removing the majority treatment we decided to exclude it from the study.⁵ For further investigation it would also be desirable to have a more detailed treatment description and potentially be able to derive a scale of “adequacy of treatment.” Additionally, the inclusion of a broader age range in future research would contribute to a more comprehensive understanding of how treatment outcomes vary across different age groups. Furthermore, given that our data had a variable component called “narrative” which briefly described the condition of the patient when they got to the ER; we analyzed the most commonly used words in the narrative by gender and race and notice a major difference. We wonder how the bias of whoever described the ailment could potentially affect the treatment received, and believe that could also be an interesting area of further study.

Conclusion

The main focus of this research was to identify and predict the sources of disparity in ER settings as part of disparities within the medical system. Even though the results require careful interpretation we succeed in finding a predictive model that displays the differences in treatment outcomes based on gender and sex. After trying three types of models and performing statistical relevance tests we were able to find evidence to answer our question⁶ with : it is likely and it contributes to disparities in diverse ways which need further research.



Graph: Multinomial logistic regression graph of p-value, based on gender and treatment.

⁵ See Appendix B

⁶ “Do race and gender contribute to disparities in ER settings? If so, in what way?”

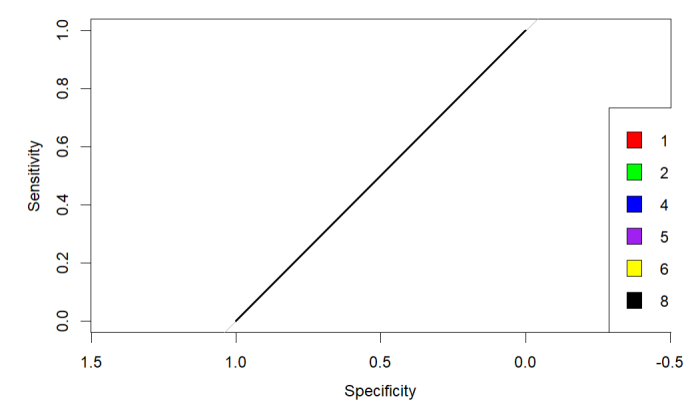
References

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- Fisher, Timothy J. "Introduction to Statistical Modeling." <https://tjfisher19.github.io/introStatModeling/>.
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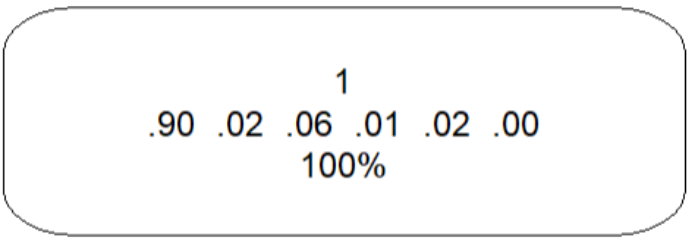
Appendix A: Visualizations within our Academic Paper

	Race0	Race2	Race3	Race4	Race5	Race6	Sex2
2	0.8989359	0.7298106	1.039774e+00	0.5729820	3.424502e-01	7.136189e-01	0.7337557
4	0.8497307	0.9755323	8.766605e-01	0.8627752	2.187080e+00	2.053388e+00	0.6296264
5	0.7051316	2.2524720	1.966886e+00	1.2816552	4.559271e+00	1.520816e-07	0.8736510
6	1.1927716	1.8597449	1.446337e+00	1.0228094	8.259953e-01	2.672383e+00	1.2110580
8	1.7809124	0.7134207	5.924258e-81	5.7423776	1.580450e-07	4.316311e-04	0.3912691

Table1: Exponent of the coefficients for "Race + Sex
* Note that there are 8 treatments in the key but the treatment no injury (treatment = 0) and treatment unknown (treatment = 9) have zero patients.



Graph 3: AUC for Random Forest



Graph 4: Decision Tree

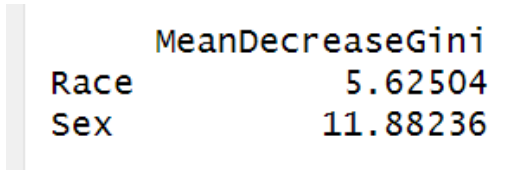
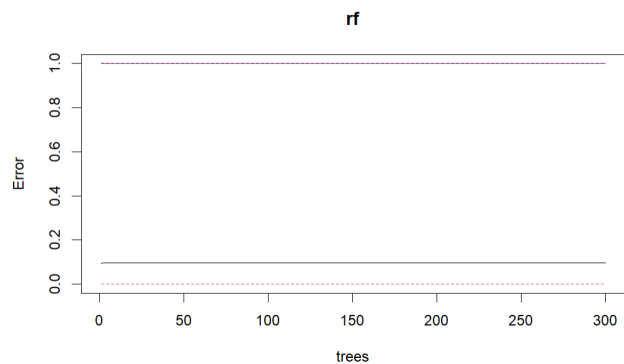
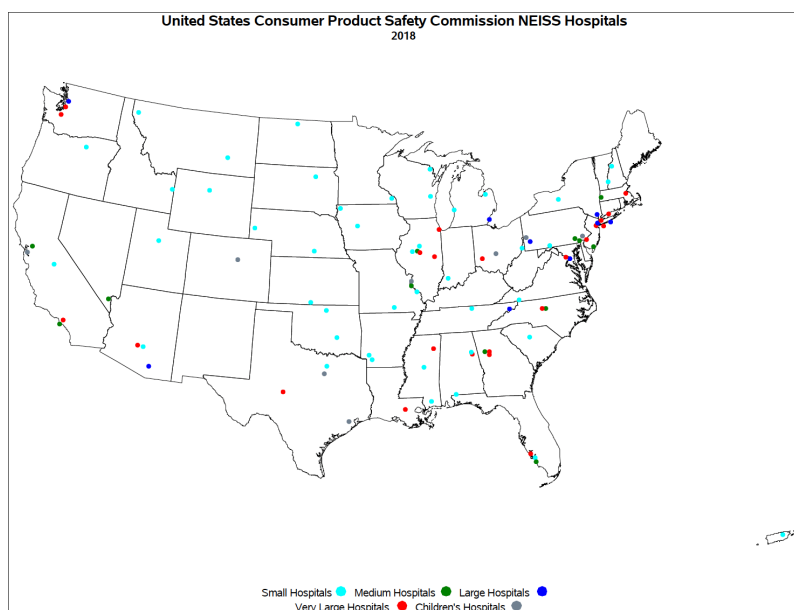


Chart 1: Random Forest Variable Importance



Graph 5: Random Forest Plot

Appendix B: Additional Info and Visualizations



Graph 1: Location of where the samples were taken in the U.S.

Looking at the most common words for each gender and sex

Looking at the most common words for each race and sex within Narrative column you can see that the terminology that patients use depending on race and gender are different. Terminology could be a factor in why there's a disparity in dispositions among genders/races.

Sex <fctr>	word <chr>	n <int>
1	SHOULDER	9326
1	DISLOCATION	7030
1	FINGER	6172
1	PLAYING	5959
1	WITH	5197
1	WHEN	4408
1	FELL	4239
1	BURN	4094
1	WHILE	3942
1	LEFT	3684

1-10 of 20 rows

Sex <fctr>	word <chr>	n <int>
2	BURN	4161
2	WITH	3022
2	FINGER	2470
2	HAND	2363
2	WHEN	2281
2	FELL	2190
2	LEFT	1952
2	PAIN	1849
2	SHOULDER	1826
2	DISLOCATION	1786

Chart 1 & 2: Most common words for Males (Male = 1) and Females (Female = 2)

Race <fctr>	word <chr>	n <int>
1	SHOULDER	4698
1	WITH	3613
1	DISLOCATION	3608
1	BURN	3558
1	FINGER	3411
1	WHEN	2892
1	FELL	2841
1	HAND	2752
1	LEFT	2732
1	PAIN	2446

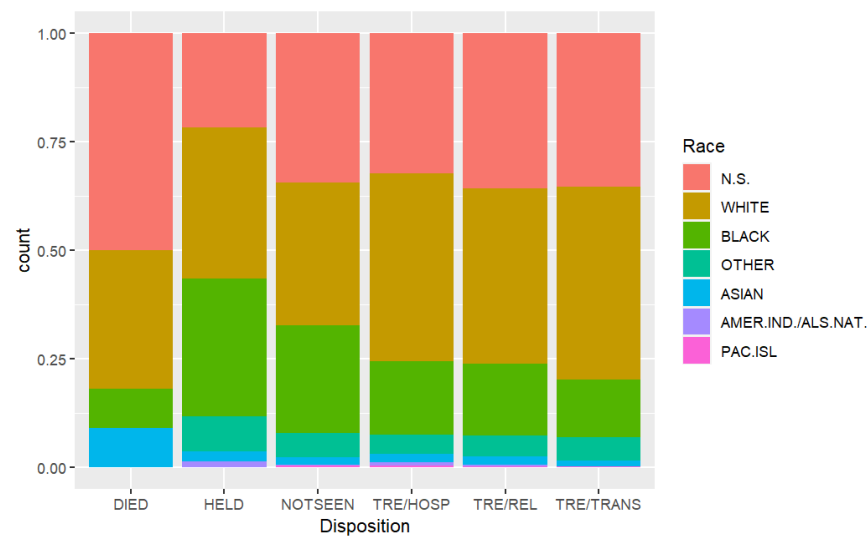
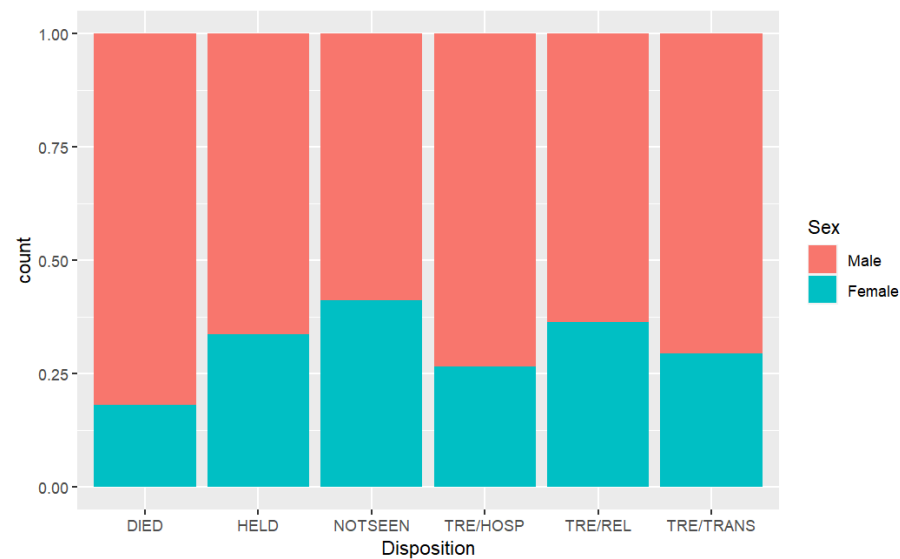
Race <fctr>	word <chr>	n <int>
6	PLAYING	25
6	FINGER	24
6	WITH	24
6	DISLOCATION	20
6	RIGHT	18
6	HAND	15
6	AFTER	14
6	FELL	14
6	PAIN	14
6	LEFT	13

Race <fctr>	word <chr>	n <int>
5	WITH	50
5	BURN	46
5	WHEN	46
5	RIGHT	37
5	SHOULDER	33
5	FINGER	29
5	HAND	29
5	FELL	26
5	KNEE	24
5	LEFT	24

Chart 3 & 4 & 5: Most common words for American Indian/Alaska Native, Native Hawaiian/Pacific Islander, and White participants

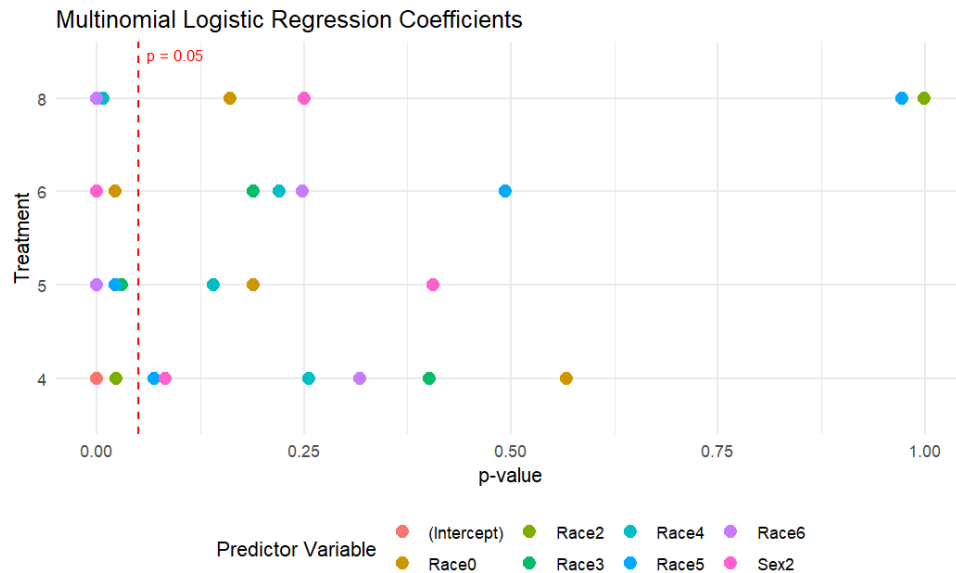
Visualization of Gender proportion and Race Proportion

Visualization of Gender proportion and Race Proportion in each Disposition/Treatment, to get a general undstanding of the data distribution



Analyzing multinomial logistic regression and results ommitting Treatment 1 from the dataset.

Treatment 1 is found to be a majority of the dataset, therefore we want to see if our predictor variables are statistically significant without the influence of Treatment 1.



It appears that by omitting Treatment 1 from the dataset that the predictor variables Race and Sex are less significant. This might suggest that there's a high amount of disparity in race/sex for treatment 1 that accounts for the overall disparity in race/sex for all treatments. It appears that omitting treatment 1 lowers the significance level of the predictor variable (females and NATIVE HAWAIIAN/PACIFIC ISLANDER) across all treatments. Further research could be done to look and see if females and NATIVE HAWAIIAN/PACIFIC ISLANDER are more fairly treated when considering various treatments, excluding the standard approach of being treated and then released. These results could have been influenced by the change in proportion of races and gender when omitting all rows with Treatment 1.